

Part A: Negative Numbers and Order of Operations

Evaluate each expression completely. Show each major order-of-operations step.

1. $10 + (-14)$

6. $-7 + (-9) \cdot (-10)$

2. $-3 + (-14)$

7. $2^2 - 17 \cdot 6$

3. $11 \cdot (-7)$

8. $9^3 - (-4) \cdot 7$

4. $8 \cdot (-2)$

9. $17 - 10 \cdot (7 + 17)$

5. $-3 + (-12) \cdot 5$

10. $(3 + 23)^2 \div 4$

Part B: Ratios, Proportions, and Percentages

Solve each problem. Show your setup and simplify your answer.

1. $\frac{7}{14} = \frac{x}{42}$

6. 60% of 600

2. $\frac{x}{11} = \frac{12}{44}$

7. from 50 to 60

3. $\frac{x}{14} = \frac{72}{112}$

8. from 5 to 3

4. 30% of 72

9. Liam pays \$9 for 3 pencils. Find the price per pencil.

5. 15% of 300

10. Kai pays \$20 for 4 bagels. Find the price per bagel.

Part C: Exponents, Roots, and Scientific Notation

Simplify or evaluate completely. Write powers with positive exponents.

1. $8x^8 \cdot 3x^9$

6. $(9x^2)^4$

2. $2x^2 \cdot 7x^6$

7. $\sqrt{49}$

3. $\frac{9x^{12}}{x^6}$

8. $\sqrt[3]{64}$

4. $\frac{6x^{11}}{x^3}$

9. 9400

10. $(4.565 \times 10^4) \div (8.3 \times 10^2)$

5. $(5x^4)^4$

Part D: Simplifying and Factoring Expressions

Simplify or factor as directed. Show distribution and like-term steps.

1. $2x + 7 + 10x + 12$

6. $10x^2 - 6x + 10x + 3x^2$

2. $-x + 3 + 2x + 5$

7. $12(2x - 4) + 12x + 10$

3. $3x - 2 - 8x + 5$

8. $11(3x - 8) + 4x - 1$

4. $-5x + x^2 - 8x - 12x^2$

9. GCF $(60x^3, 84x^1)$

5. $2x^2 - 11x^2 + 11x - 2x$

10. GCF $(24x^1, 9x^2)$

Part E: Solving Linear Equations

Solve each equation for x . Show your steps and check when practical.

1. $10x + 12 = 17$

2. $4x + 8 = 32$

3. $9(x + 9) - 12 = -3$

7. $-\frac{8}{5}x + 10 = \frac{138}{25}$

4. $2(4x - 6) + 6 = -94$

8. $\frac{x}{3} - 14 = -\frac{22}{3}$

5. $7x - 7 = 6x + 1$

9. $\frac{-3}{x} = \frac{7}{6}$

6. $-7x + 8 = 7x + 78$

10. $\frac{-3}{x} = \frac{4}{3}$

Part F: Linear Inequalities

Solve each inequality for x . Reverse the inequality when required.

1. $x - 1 \leq -12$

6. $10x - 5 > -11x + 247$

2. $x - 12 < 8$

7. $-\frac{7}{2}x + 17 \geq 52$

3. $11x - 3 \geq -14$

8. $-\frac{9}{4}x - 13 \leq -\frac{43}{4}$

4. $-4x + 1 > -19$

9. $-72 \leq -8x - 8 < 0$

5. $-12x - 11 \leq -8x - 43$

10. $-68 < -7x + 9 < 65$

Part G: Coordinate Plane and Linear Relationships

Find the requested slope, line equation, or distance.

1. $y = 7x + 6$

6. $(-9, -39)$ and $(-2, -11)$

2. $y = 9x - 7$

7. $(-3, 14)$ and $(-9, 22)$

3. $(-6, -32)$ and $(8, 31)$

8. $(-4, -7)$ and $(4, -22)$

4. $(4, 4)$ and $(6, 7)$

9. $m = 9$ through $(-8, -67)$

5. $(-5, -21)$ and $(-1, -1)$

10. $m = 6$ through $(-4, -32)$

Part H: Functions and Patterns

Evaluate, analyze, or find the requested rule or term.

1. $f(x) = x^2 + 6x + 4$, $f(-3)$

6. $\{(0, 9), (5, -2), (-1, -9), (-4, -10), (-10, 7)\}$

2. $f(x) = 2x^2 - 2x - 9$, $f(3)$

7. $\{(7, 1), (-6, -2), (-10, 2), (9, 2), (2, 6)\}$

3. $f(x) = 3x^2 - 6x + 7$, $f(4)$

8. $r = 7cs$, solve for s

4. $f(x) = 6x^2 - 2x + 9$, $f(1)$

9. $6r + 5a = q$, solve for a

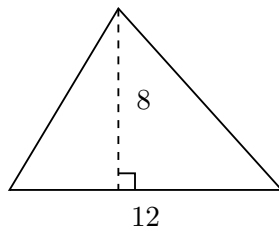
5. $\{(-8, -4), (3, -8), (10, 5), (7, 8), (-2, -10)\}$

10. $5b + 3u = a$, solve for u

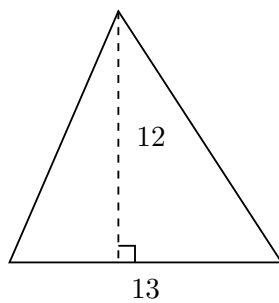
Part I: Geometry Applications

Find the requested measure. Include units when given.

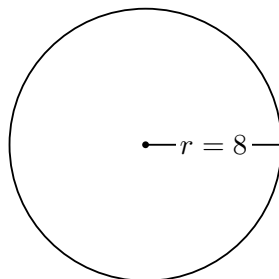
1. All measurements are in in.



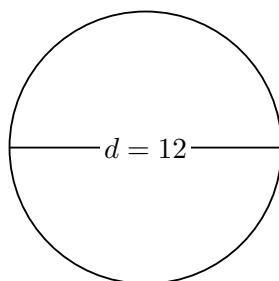
2. All measurements are in in.



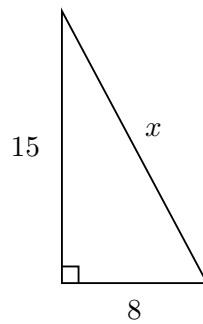
3. All measurements are in cm.



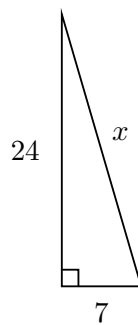
4. All measurements are in ft.



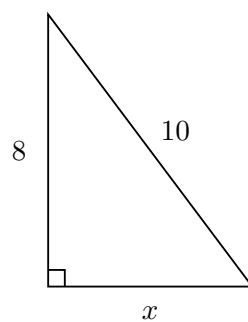
5. A right triangle. All measurements are in ft.



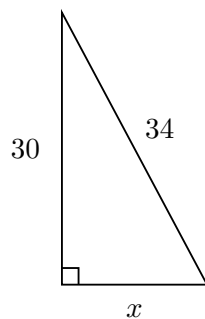
6. A right triangle. All measurements are in cm.



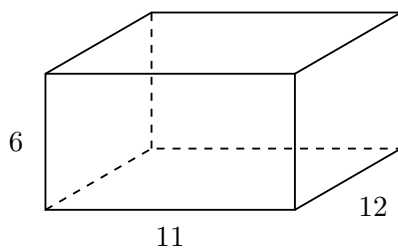
7. A right triangle. All measurements are in units.



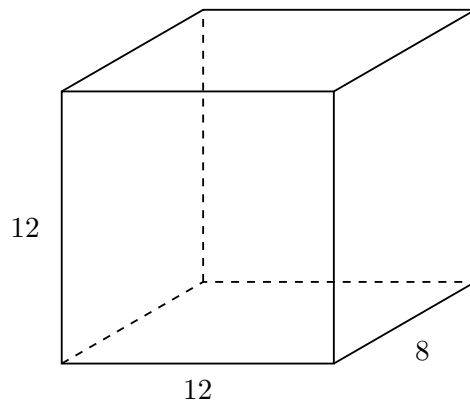
8. A right triangle. All measurements are in in.



9. A rectangular prism. All measurements are in ft.



10. A rectangular prism. All measurements are in m.



Part J: Word Problems and Modeling

Define a variable, write a model, solve, and interpret the answer.

1. Wes buys 8 marbles each week and already owns 21 marbles. After how many weeks will Wes have 117 marbles?

2. Noah buys 6 tickets each week and already owns 13 tickets. After how many weeks will Noah have 43 tickets?
3. Sam buys 5 stickers each week and already owns 2 stickers. After how many weeks will Sam have 62 stickers?
4. Priya rides at a steady 11 miles per hour and travels 66 miles. How many hours does the ride take?
5. Ethan rides at a steady 11 miles per hour and travels 33 miles. How many hours does the ride take?
6. Jonas rides at a steady 19 miles per hour and travels 95 miles. How many hours does the ride take?
7. A sticker originally costs \$80. Maya uses a 25% off coupon. How many dollars does Maya save?
8. A marble originally costs \$40. Jonas uses a 40% off coupon. How many dollars does Jonas save?
9. Plan A charges \$50 plus \$17 per month. Plan B charges \$5 plus \$22 per month. After how many months do the two plans cost the same?
10. Plan A charges \$49 plus \$9 per month. Plan B charges \$22 plus \$12 per month. After how many months do the two plans cost the same?

Answer Key

Part A: Negative Numbers and Order of Operations

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|----------|----------|-----------|
| 1. -4 | 5. -63 | 9. -223 |
| 2. -17 | 6. 83 | 10. 169 |
| 3. -77 | 7. -98 | |
| 4. -16 | 8. 757 | |

Part B: Ratios, Proportions, and Percentages

- | | | |
|--------------------|------------|-------------------|
| 1. $x = 21$ | 5. 45 | 9. \$3 per pencil |
| 2. $x = 3$ | 6. 360 | 10. \$5 per bagel |
| 3. $x = 9$ | 7. 20% | |
| 4. $\frac{108}{5}$ | 8. -40% | |

Part C: Exponents, Roots, and Scientific Notation

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|---------------|----------------|-----------------------|
| 1. $24x^{17}$ | 5. $625x^{16}$ | 9. 9.4×10^3 |
| 2. $14x^8$ | 6. $6561x^8$ | 10. 5.5×10^1 |
| 3. $9x^6$ | 7. 7 | |
| 4. $6x^8$ | 8. 4 | |

Part D: Simplifying and Factoring Expressions

- | | | |
|-------------------|-----------------|----------|
| 1. $12x + 19$ | 5. $-9x^2 + 9x$ | 9. $12x$ |
| 2. $x + 8$ | 6. $13x^2 + 4x$ | 10. $3x$ |
| 3. $3 - 5x$ | 7. $36x - 38$ | |
| 4. $-11x^2 - 13x$ | 8. $37x - 89$ | |

Part E: Solving Linear Equations

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|----------------------|-----------------------|------------------------|
| 1. $x = \frac{1}{2}$ | 5. $x = 8$ | 9. $x = -\frac{18}{7}$ |
| 2. $x = 6$ | 6. $x = -5$ | 10. $x = -\frac{9}{4}$ |
| 3. $x = -8$ | 7. $x = \frac{14}{5}$ | |
| 4. $x = -11$ | 8. $x = 20$ | |

Part F: Linear Inequalities

- | | | |
|-----------------|-----------------|--------------------|
| 1. $x \leq -11$ | 5. $x \geq 8$ | 9. $-1 < x \leq 8$ |
| 2. $x < 20$ | 6. $x > 12$ | 10. $-8 < x < 11$ |
| 3. $x \geq -1$ | 7. $x \leq -10$ | |
| 4. $x < 5$ | 8. $x \geq -1$ | |

Part G: Coordinate Plane and Linear Relationships

1. $m = 7$, $b = 6$
2. $m = 9$, $b = -7$
3. $m = \frac{9}{5}$
4. $m = \frac{3}{2}$

5. $y = 5x + 4$
6. $y = 4x - 3$
7. 10
8. 17

9. $y = 9x + 5$
10. $y = 6x - 8$

Part H: Functions and Patterns

1. -5
 2. 3
 3. 31
 4. 13
 5. domain: $\{-8, -2, 3, 7, 10\}$;
- range: $\{-10, -8, -4, 5, 8\}$
 6. domain: $\{-10, -4, -1, 0, 5\}$;
 - range: $\{-10, -9, -2, 7, 9\}$
 7. domain: $\{-10, -6, 2, 7, 9\}$;
 - range: $\{-2, 1, 2, 6\}$
8. $s = \frac{r}{7c}$
 9. $a = \frac{q - 6r}{5}$
 10. $u = \frac{a - 5b}{3}$

Part I: Geometry Applications

1. 48 square in
 2. 78 square in
 3. 64π square cm
 4. 36π square ft
5. 17 ft
 6. 25 cm
 7. 6 units
 8. 16 in
9. 792 cubic ft
 10. 1152 cubic m

Part J: Word Problems and Modeling

1. 12 weeks
 2. 5 weeks
 3. 12 weeks
 4. 6 hours
5. 3 hours
 6. 5 hours
 7. 20 dollars
 8. 16 dollars
9. 9 months
 10. 9 months